



IS231: Web Technology

Lecture 1- Course Overview



By: Neamat El Tazi

Welcome to the course

▶ Course Information

▶ Doctors:

- ▶ Dr. Mohamed Noureldin
- ▶ Dr. Neamat ElTazi

▶ TAs

- ▶ TBD

▶ Grading Scheme

- ▶ 60 % on Final Exam
- ▶ 20 % on Mid-Term Exam
- ▶ 20 % on Assignments and Project

Course Content

On Google Classroom

<https://classroom.google.com/c/NzkzNjYwOTY2ODcz?cjc=ecabhjcw>

Class Code:

ecabhjcw

Lecture Norms



Lecture Norms



No Side Talks ☺

Lecture Norms



Leaving the Lecture

Expectations



References

Books:

- **Web Application Architecture: Principles, protocols and practices By Leon Shklar and Richard Rosen**

Websites:

Websites references will be available in the slides of each lecture

Goal of the Course



The focus in this course is on the World Wide Web as a platform for interactive applications, content publishing and social services.



The development of web-based applications requires knowledge about the underlying technology and the formats and standards the web is based upon.

You'll Learn



HTTP communication protocol



Markup languages like HTML



CSS standards for formatting and transforming web content.



Client-side programming using JavaScript.



Server-side programming using Python



Front-end and Back-end frameworks

CERN

- ▶ Scientific Research Laboratory, Geneva, Switzerland
- ▶ Tim Berners-Lee presented a proposal for an information management system that would enable the sharing of knowledge and resources over a computer network.
- ▶ The system he proposed has propagated itself into what is called a World Wide Web (1989)



CERN: The origins of the web



Origin: CERN Proposal

- ▶ Tim Berners-Lee originally promoted the World Wide Web as a virtual library, a document control system for sharing information resources among researchers.
- ▶ Online documents could be accessed via a unique document address, a Universal Resource Locator (URL).
- ▶ These documents could be cross-referenced via hypertext links.

Origin: Exchanging Information

- ▶ Many people shared this dream.
- ▶ FTP and NetNews
- ▶ However, it took Tim Berners-Lee to (in his own words) “marry together” the **notion of hypertext with the power of the Internet**. His vision was to connect literally everything together, in a uniform and universal way.

Foundation: Web Protocols

- ▶ The Web is built on top of core Internet protocols that had been in existence for many years prior to the Web's inception. Understanding the relationship between 'Web technology' and the underlying Internet protocols is fundamental to the design and implementation of true 'Web applications'

Question?

**What is the difference
between a web page, a
website and a web
application????**

Origin: From a Web page to a Website

Welcome to my
homepage!
My name is Ahmed
I have a cat
I have some photos
here with my cat



Origin: From a Web page to a Website

- ▶ A Website is more than just a group of Web pages that happen to be connected to each other through hypertext links.

1. Content:

- ▶ Maintaining thematic consistency of content is important in giving a site some degree of identity.

2. Look & Feel:

- ▶ A Website should also have a common look and feel across all its pages, so that site visitors know they are looking at a particular Website. This means utilizing a common style across the site: page layout, graphic design, and typographical elements should reflect that style.

Origin: From a Web page to a Website

3. Architecture

- ▶ As a site grows and becomes more complex, it becomes critically important to organize its content properly. This includes not just the layout of content on individual pages, but also the interconnections between the pages themselves

Origin: From a Website to a Web Application

- ▶ Initially, what people shared over the Internet consisted mostly of static information found in files
- ▶ There were a few exceptions: search applications for finding files found on FTP archives
- ▶ One of the first Web applications that Tim Berners-Lee demonstrated at CERN was a gateway for looking up numbers from a phone book database using a Web browser.

Origin: From a Website to a Web Application

- ▶ **Web Application:** is a client/server application that uses a Web browser as its client program and performs an interactive service by connecting with servers over the Internet (or Intranet).
- ▶ A Web site simply delivers content from static files.
- ▶ A Web application presents dynamically tailored content based on request parameters, tracked user behaviors, and security considerations

Examples of (Dynamic Websites) Web Applications

Dynamic Websites or Web applications:

1. E-commerce. (Amazon, Souq, Jumia ...)
2. Blogs and Personal. (personal websites...)
3. Informational. (Wikipedia, CNN,...)
4. Online Community. (Facebook, Linkedin,...)
5. Photo Sharing. (Instagram)
6. Resume. (personal)
7. Portfolio. (Behance, Github...)

Course Outline

